

Scenario-based analyses of energy system development and its environmental implications in Thailand

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Abstract

Thailand is one of the fastest growing energy-intensive economies in Southeast Asia. To formulate sound energy policies in the country, it is important to understand the impact of energy use on the environment over the long-period. This study examines energy system development and its associated greenhouse gas and local air pollutant emissions under four scenarios in Thailand through the year 2050. The four scenarios involve different growth paths for economy, population, energy efficiency and penetration of renewable energy technologies. The paper assesses the changes in primary energy supply mix, sector-wise final energy demand, energy import dependency and CO₂, SO₂ and NO_x emissions under four scenarios using end-use based Asia-Pacific Integrated Assessment Model (AIM/Enduse) of Thailand.

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1. Introduction

In recent years, it has been acknowledged that adverse effects of climate change needs to be studied over a long-term horizon (UNFCCC, 2004). Signed in 1997, the Kyoto Protocol to the United Nations Framework Convention on Climate Change (UNFCCC), require reduction of greenhouse gas (GHG) emissions by industrialized countries. Developing countries (DCs) are not legally required by the protocol to reduce the GHG emissions. Since the financial crisis in 1997, Thailand has undergone a remarkable economic transformation. The country's economic growth trend has improved, from −10.2% in 1998 to 6.2% in 2004. Although economic growth in 2005 (4.5%) temporarily decelerated due to concerns over rising oil prices and short-term disturbances such as 2004 tsunami and outbreak of avian flue, the country's economic growth is projected to remain between 5% and 6% in 2006. The country is also

the second largest economy among the ASEAN¹ countries with a total gross domestic product (GDP) of US\$176 billion in 2005 and GDP per capita of US\$2824 (BOT, 2006). As a result of high economic growth, the country has also recorded strong growth in energy consumption and this growth is expected to increase as well in the future. For example, Thailand's oil consumption in 2004 was about 718,422 barrels per day that rose at an average rate of 9.7% from the previous year (DEDE, 2006a). In 2005 alone, the value of energy imports in the country accounted for 9.6% of the GDP (US\$17 billion). This increasing demand for energy would not only put pressure on the country's economy due to increased energy import dependency but it would also increase GHG and harmful local air pollutant emissions.

The existing long-term energy and emission scenarios that have been developed are either heavily aggregated at the regional levels or they are focused on the industrialized

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¹The Association of South East Asian Nations (ASEAN) includes Brunei, Cambodia, Indonesia, Laos, Malaysia, Myanmar, Philippines, Singapore, Thailand and Vietnam.

countries except in few cases.² Motivated by this fact and the rapidly changing economic and demographic structures in DCs, it is of interest to have country specific assessments of energy system development and its associated environmental implications in DCs over a long-term. There are number of studies focused on interrelationship between energy use and the environment in Thailand. Various aspects of these studies include sector specific energy planning and its impact on global and local air pollutants (see e.g., Dang et al., 1994; Shrestha et al., 1998; NEPO, 1999; Tanatvanit et al., 2003, 2004; Bhattacharyya and Ussanarassamee, 2004; Malla and Shrestha 2005; Limmeechokchai and Suksuntornsiri, 2007a, b). However, in order to formulate sound energy policies, a comprehensive understanding of national energy system development is important. In this regard, a least-cost energy system for Thailand is developed and its environmental implications are analyzed during 2000–2050 based on four scenarios. In particular, this study assesses changes in primary energy mix, sector-wise final energy demand (FED), use of renewable energy technologies, energy import dependency (EID) and sector-wise CO₂, SO₂ and NO_x emissions during 2000–2050. To assess these issues, an energy system optimization model for Thailand is developed in the framework of AIM/Enduse model—a model of the Asia-Pacific Integrated Assessment Model (AIM) (Kainuma et al., 2003). The model developed is essentially based on end-use approach that interrelate the energy service demand with the population, GDP, energy price and technological change.

The paper is divided into six sections. Section 2 describes the methodological approach. Section 3 gives a brief description of data and data sources. A detailed description of the four scenarios is discussed in Section 4. Analyses of results and key findings are presented in Section 5. The final section provides the conclusion.

2. Methodological approach

This study uses the bottom up modeling framework based on a cost minimum linear programming approach. Developed by the National Institute for Environmental Studies (NIES), Japan, AIM/Enduse model is based on the partial equilibrium framework and the model is used as a tool to estimate the future energy demand and emissions at the regional- and country-levels. It simulates the flows of energy and materials in an economy, from the source or supply of primary energy and materials, through conversion into secondary energy and materials, to the delivery of various forms of energy to the end-use services. In the model, these flows of energy and materials are character-

ized through detailed representation of technologies providing an end-use scenario-driven analysis. Fig. 1 shows the simplified structure of the AIM/Enduse modeling framework of Thailand.

The database in AIM/Enduse model includes details of energy types, energy technologies, energy-related services and counter measures. The energy database comprise of different energy types and their prices and maximum availability as its parameters while the energy technology database comprise of details corresponding to technology price, life time energy consumption per unit of output and the availability factor. Likewise, the energy service database considers the demand for each service in the economy. The model also considers the quantity of each device at the beginning of planning horizon. The options in counter measures database could incorporate environmental taxes, emission regulations or subsidies on energy or devices. Based on the energy consumption, emissions are estimated. The database comprise of input parameters that represent each stage of an energy system including primary energy production (e.g., mining), conversion of primary energy to secondary energy (e.g., electricity generation) and secondary energy consumption (e.g., petroleum products).

The linear programming framework of the model comprises an objective function to minimize total energy system cost year by year subject to a number of constraints including those on service demand, energy resource availability, existing device stock, maximum allowable quantity of devices and emissions. The total cost comprises of annualized fixed cost of recruited devices during that year, variable operating cost (operation and maintenance cost of devices, and fuel cost), cost of installing removal devices (flue gas desulfurizers for pulverized coal fired power plants, etc.) and cost of emissions taxes (carbon tax, energy tax, etc.). The formulation also provides functions to consider the existing device quantities in the starting year of the planning horizon and to calculate the retirement of the devices at the end of its life time.

In this study, the energy system of Thailand is broadly classified into two components: energy supply and conversion, and service demand. Energy supply and conversion represents energy extraction, imports and conversion of primary energy to secondary energy. In this component, coal mining, natural gas extraction, refining of crude oil and power generation is considered. For power generation, altogether twenty existing and new technologies options are considered (Appendix A). A study by NEPO (1999) is used as the basis for classification of sectors and sub-sectors of the service demand. The detailed classification of service demand, technologies and activity parameters are given in Appendix B. The activity parameters, exogenous inputs to the model, are used to estimate the future service demand.

3. Data and data sources

The data on economic and demographic parameters, service demand, technology characteristics, energy price,

²See, for example, Nakićenović et al. (2000, 2006) at the regional/global levels; MOE (2001) and Masui et al. (2006) for Japan; DTI (2003a, b) for United Kingdom; Radanne (2004) for France; Deutscher Bundestag (2002) for Germany; Shukla et al. (2004, 2006) for India; Jiang et al. (1999) and Jiang and Hu (2006) for China.

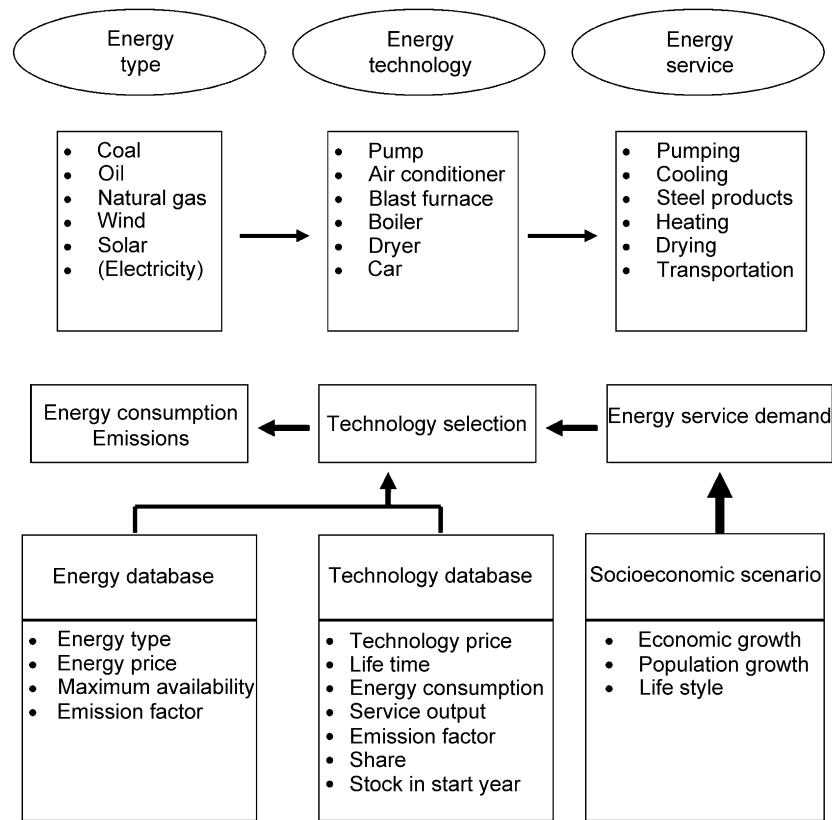


Fig. 1. Structure of the Asia-Pacific integrated assessment modeling framework of AIM/Enduse.

domestic resources availability, emission factors for CO₂, SO₂ and NO_x, appliance stock in the start year (i.e., 2000) and the maximum share of each technology type are collected from several sources and these sources are summarized in Appendix C. The data obtained from the original source are processed to meet the input requirements of AIM/Enduse framework.

Table 1 summarizes maximum availability of domestic fossil-fuel resources (coal, oil and natural gas) during 2000–2050. However, no limit is imposed on imports of these fossil-fuel resources. New and renewable energy (NRE) sources include solar, wind, hydro, geothermal, firewood, charcoal, agricultural residues, plantation-based biomass and biofuels (i.e., ethanol and biodiesel). The maximum potential of solar and wind for power generation are considered as 5000 and 1100 MW, respectively (Grea-cen, 2005). Likewise, the maximum exploitable potential of agricultural residues considered is 14,112 ktoe which comprise of sugarcane residues (i.e., bagasse and top and leaves), paddy husk, corncob and others (Santisirisomboon et al., 2003; Prasertsan and Sajjakulnukit, 2006; DEDE, 2006e). Considering the land area availability, the potential of plantation-based biomass considered is 7500 ktoe (Sajjakulnukit and Verapong, 2003; Santisirisomboon et al., 2003). The plantation-based biomass, considered only for power generation, is produced on sustainable basis and therefore the corresponding CO₂ emissions is assumed zero. Following the government's plan, the potential of

Table 1
Domestic fossil fuels resource availability in Thailand (Mtoe)

Fuel type	2005	2015	2030	2050 ^a
Coal	8.4	11.5	15.2	15.2
Crude oil	0.4	0.5	0.6	0.6
Natural gas	14.6	19.2	10.2	10.2

Source: NEPO (1999).

^a Assumed same figures as that of 2030.

biofuels considered are 3 million liters per day of ethanol and 8.5 million liters per day of biodiesel (DEDE, 2006d). The discount rate considered is 10%. All costs are expressed in constant 1995 US\$.

4. Scenarios description

The future state of the Thai economy in the long-term can be broadly visualized as a combination of market integration (i.e., extent of liberalization, globalization and the integration with the world markets) and the nature of governance (i.e., centralization vs. decentralization). These two dimensions are similar to those considered in the Intergovernmental Panel on Climate Change's Special Report on Emissions Scenarios (SRES) report and they are adapted to the Thai context (IPCC, 1996). Altogether, four scenarios are constructed. The characterization of scenarios is illustrated in Fig. 2. The following names are

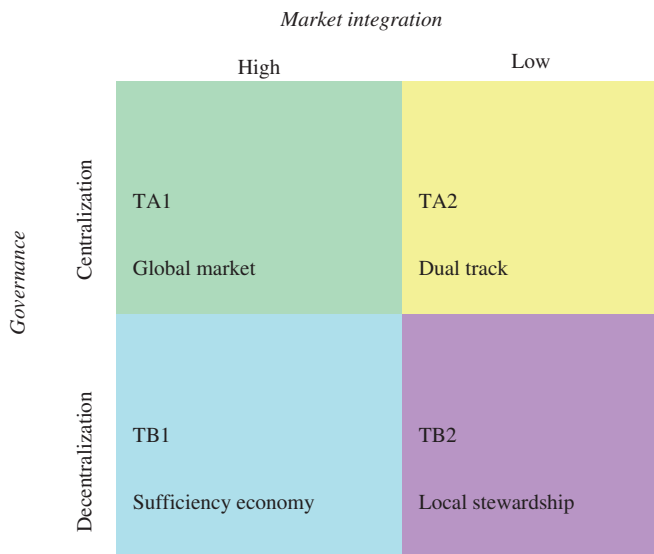


Fig. 2. Characteristics of four scenarios for Thailand.

given to four scenarios: *global market integration* (TA1); *dual track* (TA2); *sufficiency economy* (TB1); and *local stewardship* (TB2). The four scenarios can be best understood as taking a position in a combination of locus points on the two continuums: extent of market integration and the nature of governance. There may be several other scenarios on the continuum, which are also likely, but only four scenarios are constructed to cover a plausible range, in which future evolution of economy and energy is expected to lie. A brief quantitative summary of the four scenarios are presented in Appendix D.

Global market integration scenario (TA1): It is characterized by Thailand being more and more integrated into global markets. The role of market forces is predicted to strongly lead to high economic growth and there is a faster transition of the economy towards industry- and commerce-based economy. The GDP would grow by 7.5% per year during the first 20 years (2000–2020) based on the target scenario of NIDA (2006) study and by 5.5% per year for the remaining years (2021–2050) reflecting the possible slowdown of country's economic growth. The main driving force of the economic growth would be energy-intensive manufacturing of cement, steel and chemicals in the industry sector and increasing growth of commercial sector. Because of easy access to global technology market and modernization of economic sectors under this scenario, significant improvement in the efficiency of end-use devices (i.e., an average rate of 0.3% per year) is considered. In the demographic front, the country's average annual population growth is low (0.02%) due to low total fertility rate based on the United Nations low variant population projection over the study period (UN, 2004). Compared to 2000, the urbanization rate would double (62%) in 2050 due to migration from rural to urban areas and transformation of big villages into cities. The increased urbanization would lead to a rapid expansion of infrastructure, especially the transportation demand and facilities. In

terms of per capita GDP, it would increase by twenty-folds in 2050 compared to 2000 value due to both high economic growth and low population growth. Households would have high home appliances and car ownership. Demand for energy would be high in order to sustain high economic growth. The price of fossil fuels would increase by a factor of 1.5 times that of International Energy Agency's (IEA) projected annual growth over the study period (IEA, 2004b). The priority is more on the economic growth than on the conservation of environment. Overall, in this scenario, the general development mode of Thai economy is completely market-driven.

Dual track³ scenario (TA2): This scenario is characterized by the Thai economy that is concentrated on the industries that have the comparative advantage in the world market. In this scenario, Thailand follows closely the national development plans and policies. The GDP growth is moderate at an average rate of 6% per year during 2000–2020 and it would slow down to 5% per year for the remaining years until 2050. Population projection would be high at an average rate of 0.74% per year based on the United Nations high variant projection but aging population in the country would rise. In this scenario, more effective conservation measures in all economic sectors and moderate energy efficiency improvement (0.2% per year) is proposed. Households would have moderate ownership of home appliances and cars. The price of fossil fuels would increase gradually and follows IEA's projection over the study period. This scenario can be considered as a reference scenario reflecting the country's national development plan and policies. The consideration of this scenario is important to show the country's energy strategy that focuses on promotion and development of alternative and renewable energy sources, and energy management and conservation.

Sufficiency economy⁴ scenario (TB1): This scenario focuses and supports activities that promote sustainable development. GDP growth is considerable. The average annual economic growth rate would be 6.5% based on the medium scenario of NIDA (2006) study during 2000–2020 and 5.5% for the remaining years until 2050. However, the population growth would be the same as that of TA1. Clean energy and clean production would be the mainstream of the society in this scenario. Accordingly, a higher share of renewable energy technologies is introduced. The energy efficiency improvement for energy using devices

³Thai government continues to pursue the explicit policy focus on the duality of the export and the domestic sectors (dual track policy) to bring economic growth with stability. This approach represents a new policy paradigm of simultaneously pursuing the development of a strong domestic foundation for the economy as well as promoting the linkages through international trade, investment and financial cooperation (Chaipravat, 2003).

⁴The philosophy of sufficiency economy, based on adherence to the middle path, is advocated to overcome the 1997–1998 Asian economic crisis that was brought about by unexpected change under conditions of rapid globalization and achieve sustainable development (NESDB and UNRC, 2004).

would be high at 0.4% per year over the study period. The price of fossil fuels would increase by a factor of 0.5 times as that of annual growth rate of TA2. Household would have moderate appliance ownership and use of public transport is emphasized. This scenario is a consequence of active policies towards balanced economic and social differences in the country.

Local stewardship scenario (TB2): This scenario is characterized by the global economy that is unbalanced with strong economic turmoil in different regions of the world. Hence, this scenario is based on the necessity of strong local communities. Due to global turmoil, economic growth in the country is at low level, 4% per year based on low scenario of NIDA (2006) study during 2000–2020 and 3.5% per year for the remaining years until 2050. Population growth is moderate at an average 0.39% per year based on the United Nations medium variant projection. There is also low energy efficiency improvement at 0.1% per year. Household would have low home appliances ownership and high public transport use. The price of fossil fuels would remain the same as that of the current price.

5. Results and discussions

According to the formulated socio-economic and technological parameters, the Thai energy system is developed over the study period. The corresponding changes in primary energy supply mix, sector-wise FED, EID and CO₂, SO₂ and NO_x emissions under the four scenarios are presented below.

5.1. Primary energy supply

Table 2 shows the total primary energy supply (TPES) mix in 2000 and selected future years under four scenarios. Thailand is a fossil fuel intensive economy. In the year 2000, the share of fossil fuels (i.e., coal, oil and natural gas) in TPES accounted for about 81% (62 Mtoe) and the remaining share of 19% (15 Mtoe) came from NRE sources. Also in the same year, oil accounted for more than 56% of the total fossil fuels used while biomass (i.e., firewood, charcoal, agricultural residues, plantation-based

biomass and biofuels) accounted for more than 93% of the total NRE sources.

As seen from Table 2, the TPES between years 2000 and 2050 is estimated to grow from more than three-folds under TB2 to about eight-folds under TA1. Fossil fuels would continue to dominate the country's primary energy supply mix accounting for more than 86% of the TPES under all scenarios in 2050. Oil would remain a single largest fuel in the primary energy supply mix over the study period except under TA1 scenario in 2050. Among the fossil fuels use over the study period, coal would essentially be restricted to all industrial applications except fabricated metals and electricity generation, while oil would essentially be restricted to manufacturing and transportation. In the case of natural gas use, electricity generation would account for the most followed by industries (food beverages, chemicals, cement and fabricated metals production) and road transportation. In contrast, the share of NRE sources in TPES would fall from 19% in 2000 to the lowest at 6% under TA1 to about 14% under TB2 in 2050. The fall in the share of NRE sources over the study period is due to the limited domestic resource availability of biomass and limited potential of hydro, solar, wind and geothermal in the country. Renewables (i.e., biomass and hydro) would account for more than 99% of total NRE sources over the study period. New renewables (i.e., solar, wind and geothermal), however, would account for only a negligible part in TPES over the study period. By 2050, use of biofuels (i.e., ethanol and biodiesel) in transportation sector would reach about 12.7% of the total biomass requirements which corresponds to about 1% of the TPES under the reference scenario.

Thailand is a net energy import country in the past. The EID was about 50% in 2000 and it is estimated to reach ranging from 80% under TB2 to 89% under TA1 in 2050 (Fig. 3). The increase in EID over the study period is mainly due to growing energy demand and the limited domestic energy resources availability in the country. In 2000 alone, almost all oil consumption, more than 33% of total coal consumption and 10% of total natural gas consumption were imported in Thailand. Under the high growth TA1 scenario, coal and natural gas imports by 2050 are estimated to reach up to 90% and 95% of total coal

Table 2
Primary energy supply mix under four scenarios during 2000–2050 (Mtoe)

	2000	TA1			TA2			TB1			TB2		
		2010	2030	2050	2010	2030	2050	2010	2030	2050	2010	2030	2050
Coal	9	17	67	149	16	61	128	16	50	105	14	39	57
Oil	35	43	97	196	40	80	168	42	77	163	38	53	106
Gas	18	27	60	218	26	53	124	25	51	112	24	24	61
Biomass	14	21	34	34	14	34	34	19	28	28	19	24	31
Others ^a	1	1	4	4	1	3	4	1	3	3	1	3	3
Total	77	110	261	601	97	231	456	104	209	409	96	144	258

^aOthers include hydro, solar, wind and geothermal.

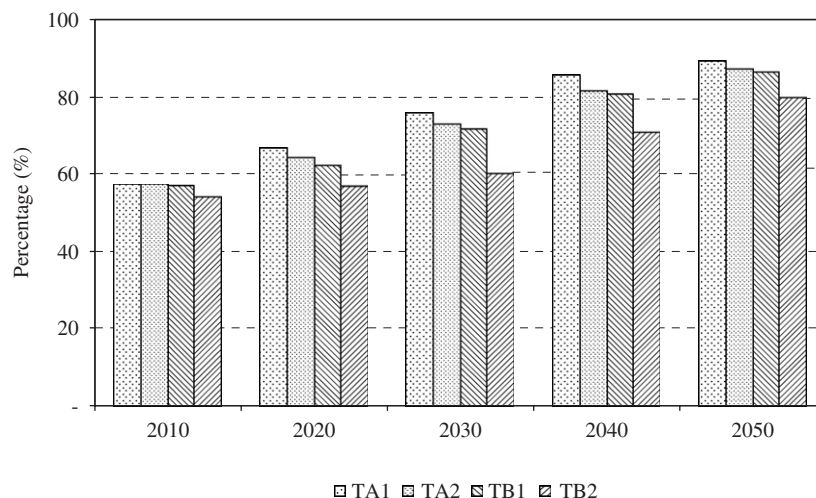


Fig. 3. Energy import dependency under four scenarios during 2010–2050 (%).

Table 3
Sector-wise final energy demand under four scenarios during 2000–2050 (Mtoe)

	2000	TA1			TA2			TB1			TB2		
		2010	2030	2050	2010	2030	2050	2010	2030	2050	2010	2030	2050
Agriculture	2	2	4	4	2	4	5	2	4	4	2	4	6
Commercial	3	5	20	56	5	17	43	5	17	43	5	10	19
Industry	18	33	88	197	31	78	156	32	78	155	28	53	83
Residential	8	10	13	17	7	13	18	9	9	11	9	10	12
Transport	18	32	71	176	29	60	114	29	56	113	25	38	81
Total	49	82	197	450	75	173	337	77	165	326	69	114	201

and natural gas requirements, respectively. The results suggest that concerns of energy security in the country would remain in decades to come.

5.2. Final energy demand

In the past, transport and industry are the two major final energy consuming sectors in the country. As seen from Table 3, these two sectors together consumed about 74% of the total FED, while the remaining 26% is consumed by residential (16%), commercial (6%) and agriculture (4%) sectors in 2000. Over the study period, the two sectors (i.e., transport and industry) together would continue to have a growing share in total FED, ranging from 81% under TB2 to 83% under TA1 in 2050. Between 2000 and 2050, the total FED is estimated to grow at an average rate of 4.5%, 4.0%, 3.9% and 2.9% per year under TA1, TA2, TB1 and TB2, respectively.

Compared to the situation in 2000, the sectoral structure in total FED is estimated to change remarkably in the future. Between years 2000 and 2050, the share of transport sector in total FED is estimated to increase from 38% to 40% under TB2, while its share is estimated to fall from 38% to 34% under TA2. On the other hand, the share of industry sector in total FED over the study period is estimated to increase

under all scenarios ranging from 41% under TB2 to 47% under TA2 in 2050 compared to 37% in 2000. Likewise, the share of commercial sector is estimated to increase by more than double (i.e., from 6% to 13%) under TA2. However, the share of residential sector in total FED is estimated to decrease from 15% in 2000 to 3% in 2050 under TB1. The decrease in the share of residential sector in total FED over the study period is mainly due to slower population growth and the use of more efficient energy consuming appliances. Likewise, the share of agriculture sector in total FED is estimated to decrease from 4% in 2000 to 1% in 2050 under TA1, TA2 and TB1. Although the share of commercial sector in total FED is relatively low in absolute value, its demand would grow most rapidly, at an average annual rate of 5.5% during 2000–2050 under the reference scenario, followed by industry (4.4%), transport (3.7%), residential (1.8%) and agriculture (1.8%) sectors. The results suggest that Thai economy would remain energy intensive over the study period.

5.3. Electricity generation and demand

Due to limitation of space, in the following, only the results of reference scenario are highlighted as it represents the most likely scenario in Thailand. Over the past 30

years, demand for electricity grew much faster than Thai economy. The Thai economy grew at an average rate of 3.7% per year, while the total electricity generation grew at an average rate of 10.3% per year during 1971–2002 (IEA, 2004c). Nonetheless, the total electricity generation over the study period is estimated to increase at much lower rate (4.4% per year), i.e., from 92 TWh in 2000 to 793 TWh in 2050 (Fig. 4). In terms of electricity generation mix, coal- and gas-fired generation would provide 85% of total electricity generation in 2050, while the remaining 15% of the electricity generation would come from NRE sources. The limited role of NRE sources in total electricity generation over the study period is due to barriers of high investment requirements and their limited domestic resource availability in the country.

The contribution of natural gas in total electricity generation is estimated to decrease from 60% in 2000 to about 43% in 2050 but remains the mainstay of electricity generation. In contrast, the corresponding figure for coal increases steadily from 21% in 2000 to about 42% in 2050. The increasing share of coal over natural gas in total electricity generation is mainly due to relatively cheaper price of coal as compared to price of natural gas and the limitation of domestic natural gas resource availability in the country. The share of oil, already low (10% in 2000), is estimated to decline further during 2000–2050. Among the share of NRE sources in total electricity generation in year

2000, hydro and new renewables together accounted for about 10% and there was no biomass-based electricity generation in the country. However, the share of NRE sources in total electricity generation would increase to 15% in 2050 and this share comprises of biomass (8%), hydro (4%) and new renewables (3%).

Sector-wise, the average annual growth of electricity demand would be the highest in transport sector (6.4%), followed by commercial (5.1%), industry (4.3%), residential (3.5%) and agriculture (1.6%) sectors (Table 4). Although the share of transport in total electricity demand is small in absolute value, the highest growth of electricity demand in transport sector is due to Thailand's plan of extending the current electricity operated mass transit rail system from around Bangkok city area to Greater Bangkok area. Industry sector is estimated to remain the largest final consumer of electricity during 2000–2040, with its consumption increasing from 39 TWh (45%) in 2000 to 247 TWh (44%) in 2040. In 2050, however, the share of commercial sector in total electricity demand is estimated to be the largest (44%), followed by industry (42%) and residential (14%) sectors.

5.4. Technology deployment in road transportation

In the past, gasoline and diesel have become the dominant fuels used in the road transport sector in the

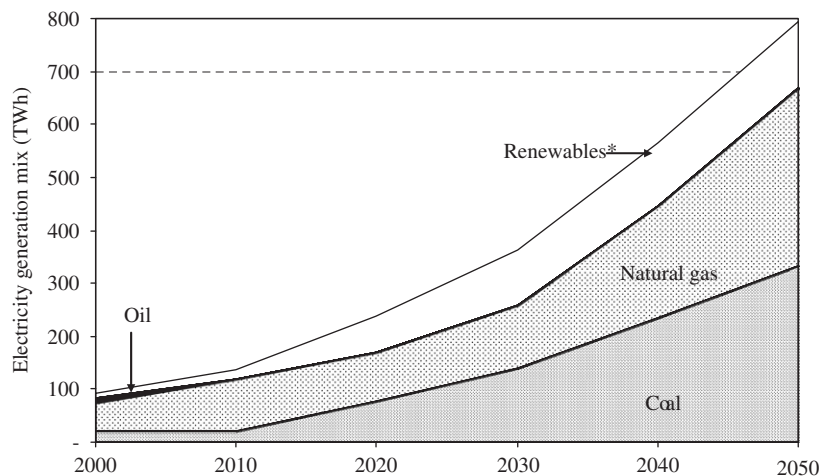


Fig. 4. Electricity generation mix under the reference scenario (TA2) during 2000–2050 (TWh). Note: *Renewables include biomass, hydro, solar, wind and geothermal.

Table 4
Sector-wise electricity demand under the reference scenario (TA2) during 2000–2050 (TWh)

	2000	2010	2020	2030	2040	2050	Average annual growth rate (%)
Agriculture	0.3	0.3	0.4	0.5	0.6	0.6	1.6
Commercial	27.9	44.4	92.6	135.6	236.7	332.9	5.1
Industry	39.3	64.9	119.6	169.5	246.8	318.6	4.3
Residential	18.9	22.6	35.1	47.0	77.2	107.4	3.5
Transport	<0.0	0.1	0.2	0.4	0.5	0.8	6.4
Total	86.4	132.3	247.9	353.0	561.7	760.3	4.4

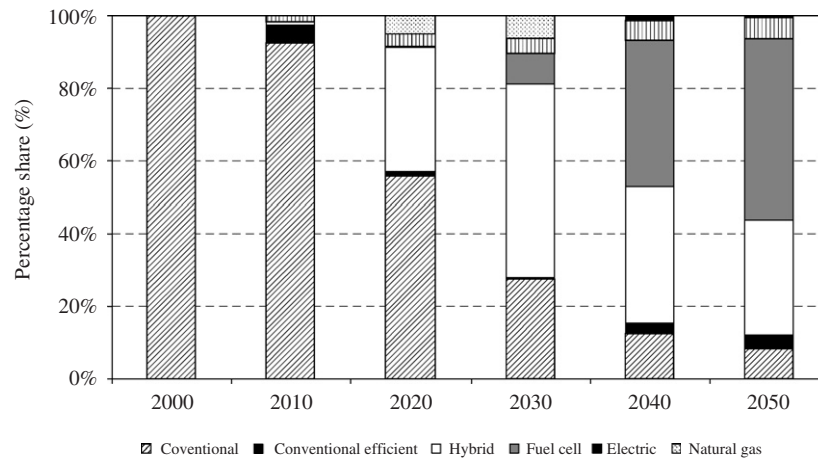


Fig. 5. Shares of existing and new technologies in road transport sector under the reference scenario (TA2) during 2000–2050 (%). *Note:* Conventional technologies include gasoline-, diesel- and LPG-cars and pickups, gasoline- and diesel-vans, diesel buses and trucks, and two wheelers; Conventional efficient technologies include new diesel car, direct ignition gasoline car, new diesel truck, new diesel air conditioned bus, new two wheeler, new diesel van and new diesel pickup.

country. However, increasing international fuel price, limited domestic availability of these fuels and the concerns of cleaner environment in the country have prompted consideration of various alternative technologies (e.g., biofuels, hybrid engines, natural gas, fuel cells and electric vehicles). Indeed, although relatively small in the share, biofuels- and natural gas-based vehicles are already introduced in the country. Fig. 5 depicts the relative share of penetration of conventional, efficient conventional and new vehicle technologies during 2000–2050 under the reference scenario. As seen from Fig. 5, the penetration of conventional efficient vehicles is estimated to be about 4.8% of total vehicle technology types in 2010 and gradually decreases thereafter. Among the technologies, the share of hybrid and fuel cell vehicle stocks together in total vehicles types is estimated to reach 81% in 2050, followed by conventional vehicle stocks (8%), electric (6%), conventional efficient (4%) and natural gas vehicles (0.4%). Over the study period, the results show that conventional vehicles would gradually be replaced by efficient conventional vehicles and these vehicles in turn would be gradually replaced by new vehicle technologies such as hybrid and fuel cell vehicles.

5.5. Environmental effects of the energy and emission scenarios

The country's CO₂ emissions in year 2000 corresponds to only about 0.7% of the world's total energy-related CO₂ emissions and about 7% of the corresponding figure of Asia excluding China (IEA, 2004a). However, the estimated trends of energy use under four scenarios imply that energy-related CO₂, SO₂ and NO_x emissions would increase over the study period. As seen from Fig. 6, the total CO₂ emissions in the country vary significantly across the scenarios, i.e., from 158 Mt in 2000 to 647 Mt under TB2 and 1312 Mt under TA1 in 2050. In the reference

scenario, the total CO₂ emissions is estimated to reach 1155 Mt in 2050, an increase of 997 Mt over the 2000 level at an average annual growth rate of 4.1%.

In conformity with the global trends, the CO₂ emissions in Thailand from the power, industry and transport sectors would have the strongest tendency to increase in the future (Fig. 7). Under the reference scenario, the three sectors together would contribute more than 94% of the total CO₂ emissions in 2050. The corresponding figures for the other three scenarios also show the similar trends varying from 93% under TB2 to 96% under TB1. In the reference scenario, industry sector would account for 38% of the increase in CO₂ emissions during 2000–2050, power sector for 33%, transport sector for 23% and the remaining sectors (i.e., agriculture, commercial and residential) for 6%.

A simple decomposition analysis⁵ reveals the important effects of energy use in the increase of CO₂ emissions in the country over the study period under the reference scenario (Fig. 8). In Fig. 8, the changes in CO₂ emissions are shown in comparison with the benchmark year 2000. The total

⁵In the literature, different decomposition techniques are available and these techniques are broadly grouped into those based on the Divisia index and those based on the Laspeyres index. The method applied in this study is a simple complete decomposition based on the Laspeyres index (Sun, 1996). The factors behind the change in CO₂ emissions from fuel combustion is based on the partition presented in the following equation:

$$CO_2 = \left(\frac{CO_2}{TPES} \right) \times \left(\frac{TPES}{FED} \right) \times \left(\frac{FED}{GDP} \right) \times \left(\frac{GDP}{POP} \right) \times POP.$$

The first factor ($CO_2/TPES$) in the equation refers to CO₂ emission intensity of entire energy system; the second factor ($TPES/FED$) refers to efficiency of energy transformation system; the third factor (FED/GDP) refers to energy intensity; the fourth factor (GDP/POP) refers to per capita GDP; and the fifth factor (POP) refers to population. For different decomposition techniques used in energy modeling studies see Ang (2004), Ang and Zhang (2000), and Rose and Casler (1996).

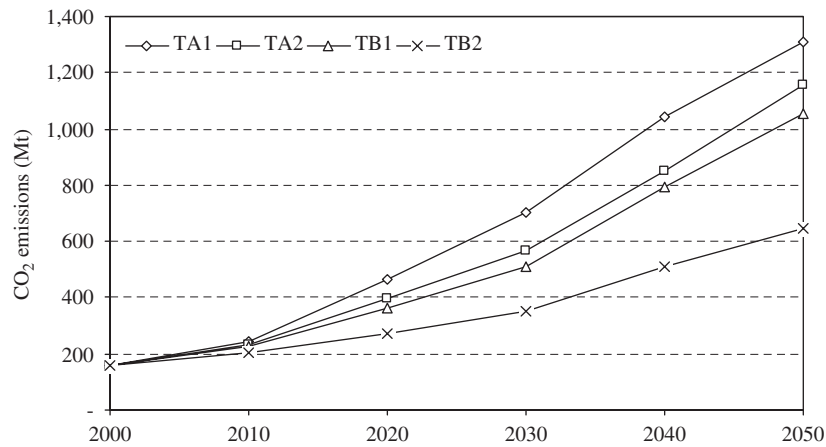


Fig. 6. Total CO₂ emissions under four scenarios during 2000–2050 (Mt).

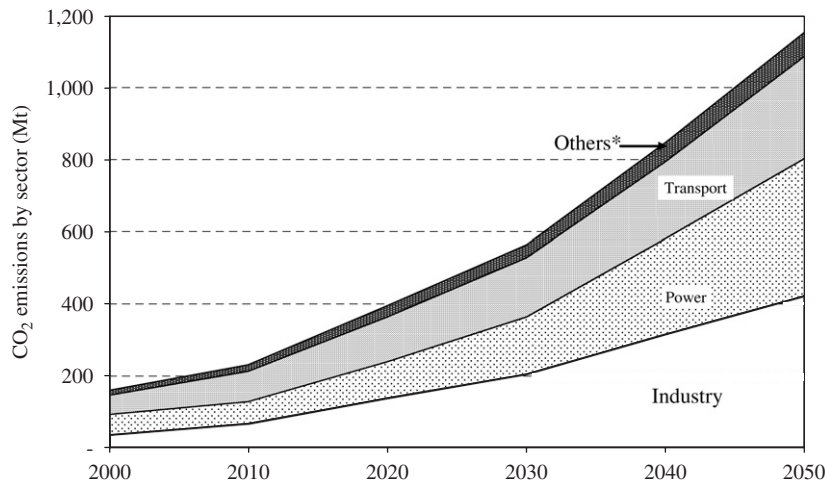


Fig. 7. Sector-wise CO₂ emissions under the reference scenario (TA2) during 2000–2050 (Mt). Note: *Others include agriculture, commercial and residential sectors.

change in the country's CO₂ emissions would be 792% increase between 2000 and 2050. Among the different factors, fuel shift (CO₂/TPES) shows that country would continue to have fossil fuels dominated energy system that leads to about 20% increase in CO₂ emissions in 2050 compared to the level of 2000. On the other hand, the efficiency increase gained by energy transformation (TPES/FED) would decrease CO₂ emissions by 7%. The modest change in energy intensity (FED/GDP) due to structural shift in the economy would decrease CO₂ emissions by 49% while the modest increase of population growth (POP) would increase CO₂ emissions by 47%. However, the per capita economic growth (GDP/POP) would be the most significant factor in the increase of CO₂ emissions and this factor accounts for over 782% increase in the total CO₂ emissions.

As seen from Table 5, the CO₂ emissions per capita is estimated to increase between years 2000 and 2050 ranging from about 3.5 times under TB2 to about 10 times under TA1. In contrast, the CO₂ emissions intensity of 0.91 kg/US\$, 1995 price under market exchange rate (MER) in year

2000 is estimated to decrease by almost half under TB1 (i.e., 0.43 kg/US\$, 1995 price using MER) and by about one-fourth under TB2 (i.e., 0.63 kg/US\$, 1995 price) in 2050. Likewise, the primary energy intensity of 0.44 kgoe/US\$, 1995 price using MER in year 2000 is estimated to decrease by one-third under TB1 (i.e., 0.14 kgoe/US\$, 1995 price using MER) and by more than half under TA1 (i.e., 0.23 kgoe/US\$, 1995 price using MER) in 2050.

The significant increase in energy demand over the study period also induces substantial increase in emissions of local pollutants in the country. Table 6 summarizes the results of total SO₂ and NO_x emissions under four scenarios during 2000–2050. Both SO₂ and NO_x emissions are estimated to increase considerably in the future: SO₂ emission is estimated to increase by about 5.3 times under TB2 and by about 9.2 times under TA1; while NO_x emission is estimated to increase by 3.6 times under TB2 and by 6.2 times under TA1 in 2050 compared to year 2000. The results also indicate that increase in SO₂ emissions is relatively higher compared to increase in NO_x emissions during 2000–2050 under all four scenarios.

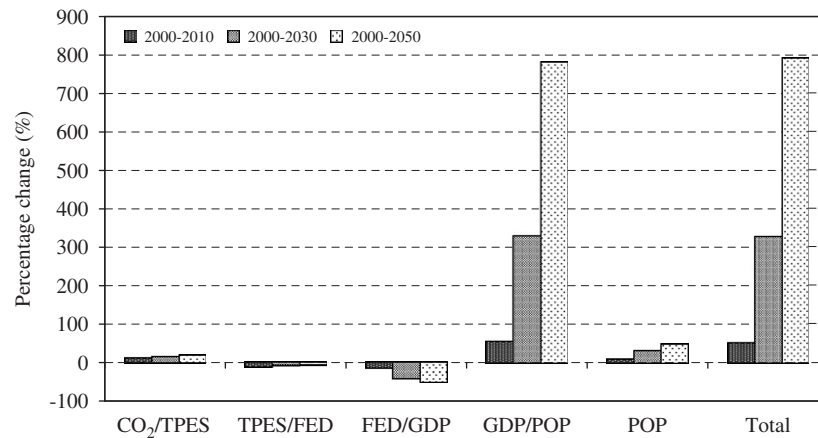


Fig. 8. Decomposition analysis for the factors affecting CO₂ emissions under the reference scenario (TA2) during 2000–2050 (%). *Note:* CO₂: carbon dioxide; TPES: total primary energy supply; FED: final energy demand; GDP: gross domestic product; and POP: population.

Table 5

CO₂ emissions per capita, CO₂ emissions intensity and primary energy intensity in 2000 and selected future years under four scenarios

	2000	Global market		Dual track		Sufficiency economy		Local stewardship	
		2030	2050	2030	2050	2030	2050	2030	2050
Per capita CO ₂ emission (tons/capita)	2.56	9.70	25.43	7.26	14.12	7.51	17.94	4.92	8.99
CO ₂ emission intensity (kg/US\$,1995 price)	0.91	0.55	0.46	0.60	0.57	0.50	0.43	0.64	0.63
Primary energy intensity (kgoe/US\$,1995 price)	0.44	0.20	0.15	0.23	0.19	0.18	0.14	0.27	0.23

Table 6

SO₂ and NO_x (as NO₂) emissions under four scenarios during 2000–2050 (kt)

	SO ₂ emissions				Ratio 2050/2000	NO _x emissions				Ratio 2050/2000
	2000	2010	2030	2050		2000	2010	2030	2050	
TA1	899	1324	5813	8288	9.2	926	1510	3307	5772	6.2
TA2	899	1314	5283	7061	7.9	926	1380	3003	5283	5.7
TB1	899	1112	4221	6798	7.6	926	1281	2795	5023	5.4
TB2	899	979	4015	4768	5.3	926	1224	1978	3375	3.6

This is mainly due to increasing use of coal in electricity generation over the study period.

Sector-wise, the power sector would contribute the most (58%) in the total SO₂ emissions in 2050 under the reference scenario, followed by industry (28%), transport (12%) and others (2%) (Fig. 9). In contrast, the transport sector would contribute the most (49%) in total NO_x emissions in 2050 under the reference scenario with the remaining emissions coming from power (23%), industry (23%) and others (6%) (Fig. 10).

6. Concluding remarks

This paper examined the least-cost energy system development and analyzed its implications for GHG and other local pollutant emissions under four scenarios for

Thailand using AIM/Enduse model during 2000–2050. The four scenarios follow closely with the scenarios of global IPCC SRES report. Under the reference scenario (i.e., dual track), the TPES is estimated to increase by almost six-folds from 77 Mtoe in 2000 to 456 Mtoe in 2050. Fossil fuels would continue to remain the dominant energy source, contributing about 90% of the total TPES under all four scenarios in 2050. Among the fossil fuels, oil use is estimated to dominate the TPES over the study period. Coal and natural gas are also estimated to remain the mainstay of TPES over the study period. Despite the Thai government's policy to promote new and renewable energy sources, its share in TPES is estimated to remain low in between 6% under TA1 to 13% under TB2 in 2050. This is mainly due to higher initial costs and lower plant factor of renewable energy technologies used for electricity genera-

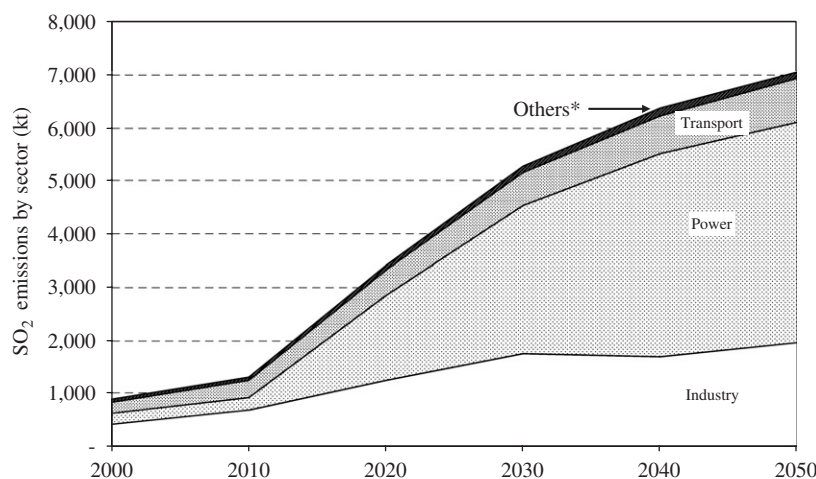


Fig. 9. Sector-wise SO_2 emissions under the reference scenario (TA2) during 2000–2050 (kt). Note: *Others include agriculture, commercial and residential sectors.

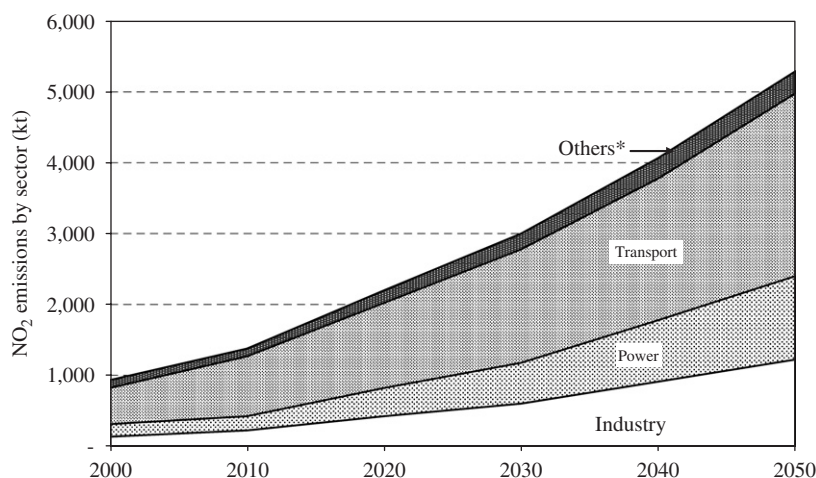


Fig. 10. Sector-wise NO_x emissions under the reference scenario (TA2) during 2000–2050 (kt). Note: *Others include agriculture, commercial and residential sectors.

tion, and limited domestic resource availability of the new and renewable energy sources in the country. Sector-wise, energy demand in industry sector would increase by almost nine-folds from 18 Mtoe in 2000 to 156 Mtoe in 2050, while transport sector would increase by more than six-folds from 18 Mtoe in 2000 to 114 Mtoe in 2050 under the reference scenario. Industry and commercial sectors are likely to remain the largest final consumer of electricity throughout the study period. In the power sector, the share of coal and natural gas combined in total electricity generation under the reference scenario is estimated to account for 85% by 2050. In contrast, the share of new and renewables (mainly biomass and hydro) in total electricity generation is estimated to increase from 9% in 2000 to about 15% in 2050. Energy use in the road transportation is estimated to become increasingly important over the study period compared to other modes of transportation (e.g., rail, air and water).

Compressed natural gas-, hybrid-, fuel cell- and other efficient technology based-vehicles would play an increasingly important role during the later two decades of the study period. Thailand is a net energy importing country and its energy import dependency is estimated to increase from 50% in 2000 to as high as 89% under global market scenario in 2050. With increasing use of fossil fuels and limited domestic resources availability, the concern over energy security would be an important issue in the country during the study period.

Owing to increasing demand for energy and increasing motor vehicles in the country, total CO_2 emissions is also estimated to increase in the future. Under the reference scenario, the total CO_2 emissions is estimated to reach 1155 Mt in 2050, an increase of more than seven-folds over the 2000 level at an average annual growth rate of 4.1%. Industry sector would account for the most in total CO_2

emissions in 2050 at 38%, followed by power (33%), transport (23%) and the agriculture, commercial and residential sectors combined (6%). In per capita term, the CO₂ emission in Thailand is estimated to increase by more than five-folds from 2.6 tons in 2000 to 14.1 tons by 2050. In contrast, the CO₂ emission intensity is estimated to fall by one-third in 2050 as compared to 2000 level due to cleaner energy substitution and penetration of more efficient technologies over the study period. However, the country's estimated CO₂ intensity in 2050 would remain high compared to the corresponding 2000 value of OECD- and world-average. Both SO₂ and NO_x emissions are also estimated to increase considerably over the study period: SO₂ emission is estimated to increase by almost eight-folds while NO_x emission is estimated to increase by almost six-folds between 2000 and 2050 under the reference scenario. In 2050, about 86% of the total SO₂ emissions would come from power and industry sectors combined, while 49% of the total NO_x emissions would come from the transport sector alone. In all four scenarios, SO₂ emission is estimated to grow much faster than the NO_x emission over the study period mainly due to the substantial use of coal for electricity generation.

In general, bottom up approach includes detailed sectoral analyses but includes much less detail about the

entire economy. Conversely, top down approach provide little sectoral details. Therefore, linking bottom up approach with top down approach for scenario analysis of developing countries like Thailand would be an interesting further study.

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Appendix A. Existing and candidate power generation technologies considered in the study

Technology	Fuel type
A. Fossil fuel and biomass-based technologies	
Conventional steam	Lignite, natural gas, fuel oil, biomass
Integrated gasification combined cycle (IGCC)	Lignite and bituminous coal
Pressurized fluidized bed combustion (PFBC)	Lignite and bituminous coal
Combined cycle	Natural gas and fuel oil
Combined cycle—advanced	Natural gas
Gas turbine	Natural gas and fuel oil
Biomass integrated gasification combined cycle (BIGCC)	Biomass
B. Renewables-based technologies	
Hydro, wind, solar photovoltaic, solar thermal and geothermal	—
C. Proton exchange membrane fuel cell	
	Natural gas

Appendix B. Classification of service demand component of energy system development of Thailand

Sector/sub-sector	Energy service	Number of technology options	Activity parameters
Agriculture	Pumping, tilling	4	Agricultural sector value added
Commercial	Air-conditioning, lighting, thermal use	9	
Industry			
Cement	Mixing, clinker production, milling, others	11	Value added for cement

Steel	Primary steel production (coke, sinter, pig iron, crude and rolled steel production) Secondary steel production (crude steel and finished steel production)	44	Value added for steel
Sugar	Juice extraction, slurry production, crystallization and grinding	9	Value added for food
Paper	Pulp production, bleaching, stock production and others	11	Value added for paper
Others (chemicals, food and beverages, fabricated metals, textiles, others)	Thermal and mechanical energy	72	Value added of the respective sub-sectors
Residential			
Metropolitan Electricity Authority (MEA)/Provincial Electricity Authority (PEA)	Air-conditioning, cooking, lighting, fan, iron, refrigerator, television, other	32	Number of households, appliance ownership per household ^a
Transport			
Passenger	Road (passenger car, micro bus and pickup, van and pickup, three wheeler, urban taxi, fixed route taxi, motor tricycle taxi, motorcycle, fixed route bus, bus for hire, private bus, small rural bus, urban train)	79	Per capita GDP, average distance traveled by each vehicle, occupancy factor ^b
Freight	Rail and air Road and water	9 12	Per capita GDP Transport sector value added

^aAppliance ownership is the number of houses that own a specific appliance as a fraction of total number of houses.

^bOccupancy factor is the average number of persons per vehicle at a given time of travel.

Appendix C. List of sources used in Thai energy system development

Category	Source
Economic	NESDB (2003), NIDA (2006)
Demographic	UN (2004)
Service demand	KMIT (1997), KMUTT (2003a, b), NEPO (1999), NSO (2003), IEA (2005); OCLMT (2000)
Technology type	Kainuma et al. (2003), NEPO (1999), Tseng et al. (2005)
Energy	DEDE (2004, 2006a–c), EGAT (2004), Greacen (2005), IEA (2004b), Prasertsan and Sajjakulnukit (2006), Sajjakulnukit and Verapong (2003), Santisirisomboon et al. (2003)
Emissions factors	IPCC (1996), Foell et al. (1995)
Appliance stock in the start year	NEPO (1999), KMUTT (2003a, b), DLT (2000)

Appendix D. Summary of main features of four scenarios

Scenario	Main features
TA1	Economy: GDP growth rate of 7.5% per year during 2000–2020 and 5.5% per year during 2021–2050 Demography: low population growth rate of 0.02% per year Life style: high per capita appliance ownership in households, car ownership and mobility Technology: energy efficiency improvement of 0.3% per year

TA2	Economy: GDP growth rate of 6% per year during 2000–2020 and 5% per year during 2021–2050 Demography: high population growth rate of 0.74% per year Life style: medium per capita appliance ownership in households Technology: energy efficiency improvement of 0.2% per year
TB1	Economy: GDP growth rate of 6.5% per year during 2000–2020 and 5.5% per year during 2021–2050 Demography: low population growth rate of 0.02% per year Life style: medium per capita appliance ownership in households and use of public transport Technology: energy efficiency improvement of 0.4% per year
TB2	Economy: GDP growth rate of 4% per year during 2000–2020 and 3.5% per year during 2021–2050 Demography: medium population growth rate of 0.39% per year Life style: low per capita appliance ownership in households and use of more public transport Technology: energy efficiency improvement of 0.1% per year

Appendix E. Supplementary data

Supplementary data associated with this article can be found in the online version of 10.1016/j.enpol.2006.11.007.

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